Free Model Stack — Model Guide

27 keyless models across the free frontier lane, the Kilo pool and failover combos — ranked for practical use in OpenCode

Generated 2026-11-03 · provider **00-free** (127.0.0.1:18900) · roster is live, re-run `npm run sync` after it changes · ids and numbers match the `/models` picker

1 How to read this

Every model below is served by the local **Our Free Model** gateway with **no login and no API key**. The rank is the picker order: a practical blend of **capability ceiling, context/output size and measured latency** — it is guidance for picking a model fast, not a benchmark result. Free lanes rotate: when one model returns 429/503, the combo or a router covers the same request.

> **Lane meaning.** `free` = OpenCode Zen free lane (frontier models, large injected system prompt ≈ 5.6–6.4k tokens per request). `kilo` = Kilo AI's keyless *isFree* pool (≈2k tokens overhead; **prompts may be logged upstream**). `combo` = a local failover chain, not a model. **Vision** = accepts images. **Out** = max output tokens — the column that quietly matters most for big rewrites and document generation.

2 Ranked roster (27 models)

#	MODEL	LANE	CONTEXT	OUT	VISION	SPECIALIZES IN
00	**Auto · recommended-free combo** combo:recommended-free	combo	1M	131k	✓	Safest default: falls over to the next free link if one dies before answering. Use when a specific model keeps 429ing.
01	**DeepSeek V4.1 Flash · free** deepseek-v4.1-flash-free	free	1M	384k	✓	All-round flagship. Strong reasoning and agentic coding, enormous output budget — the default daily driver.
02	**GLM 5.3 Flash · free** glm-5.3-flash-free	free	1M	131k	✓	Fast general reasoning, strong multilingual work, vision included — great second opinion to 01.
03	**MiMo V2.6 Flash · free** mimo-v2.6-flash-free	free	1M	131k	✓	Balanced multimodal generalist; smooth on mixed text + image conversations.
04	**Nemotron 3.5 Super VL · free** nemotron-3.5-super-vl-free	free	1M	131k	✓	Vision-language specialist: screenshots, UI review, charts, dense OCR-style reading.
05	**Nemotron 3.5 Lightning · free** nemotron-3.5-lightning-free	free	128k	32k	—	Latency-first reasoning; snappy chat and tight tool loops where 128k context is enough.
06	**Kilo Auto · free router** kilo-auto/free	kilo	256k	32k	—	Kilo's own task router across its free pool — good hands-off option, cheapest overhead on the Kilo side.
07	**Nemotron 3 Ultra 550B · kilo** nvidia/nemotron-3-ultra-550b-a55b:free	kilo	1M	65k	—	Biggest brain in the stack: hardest reasoning, deep analysis, long research tasks.
08	**Nemotron 3.5 Super VL · kilo** nvidia/nemotron-3.5-super-vl:free	kilo	1M	131k	✓	Vision workhorse with 1M context: diagram-to-code, screenshot debugging, chart extraction.
09	**Nemotron 3.5 Lightning · kilo** nvidia/nemotron-3.5-lightning:free	kilo	1M	65k	—	Speed + 1M context: fast triage of very long logs, docs and diffs.

#	MODEL	LANE	CONTEXT	OUT	VISION	SPECIALIZES IN
10	**Inkling · kilo** thinkingmachines/inkling:free	kilo	1M	262k	✓	Agentic reasoning with vision and big outputs; built for multi-step tool work.
11	**Inkling Small · kilo** thinkingmachines/inkling-small:free	kilo	1M	262k	✓	Cheaper Inkling: same 1M/262k envelope, ideal for repeated agent loops.
12	**Dots3 Note Preview · kilo** dots-studio/dots-3-note-preview:free	kilo	512k	460k	✓	Document-scale tasks: whole repo/book in, book-length output. Rare 460k output ceiling.
13	**Glyph Cluster · kilo** stealth/glyph-cluster	kilo	256k	256k	—	Massive transforms: full rewrites, exhaustive summaries, big translation passes.
14	**Flint 1.0 · kilo** recursive/flint-v1.0:free	kilo	262k	236k	—	Open reasoning model with long outputs; solid second choice for large edits.
15	**North Mini Code · kilo** cohere/north-mini-code:free	kilo	256k	64k	—	Compact coding specialist: refactors, code review, tight diff generation.
16	**OpenRouter Free Router · kilo** openrouter/free	kilo	200k	32k	✓	Routes across OpenRouter's own `:free` catalog — variety when Kilo's pool is busy.
17	**Gemma 4 31B · kilo** google/gemma-4-31b-it:free	kilo	262k	32k	✓	Small vision generalist; strong multilingual instruction following.
18	**Big Pickle · free** big-pickle	free	262k	131k	✓	General chat with vision and a roomy output budget; comfortable everyday fallback.
19	**Beam · free** beam-free	free	131k	32k	—	General reasoning, mid-size; dependable mid-tier when the flagships queue up.
20	**Beam 1.0 · kilo** reflection/beam-free	kilo	1M	131k	—	Reasoning at 1M context — long analysis without the free-lane overhead.
21	**Flint 40B · free** flint-40b-free	free	131k	32k	—	40B open reasoning, solid mid-tier judgment on everyday engineering questions.
22	**Pulse Alpha · free** pulse-alpha-free	free	131k	32k	—	Experimental generalist — a taste of upcoming free frontier models; expect volatility.
23	**LFM2.5 2.6B · kilo** liquid/lfm-2.5-2.6b:free	kilo	64k	8k	—	Tiny and fast: classification, extraction, titles — for when every millisecond counts.
24	**Xing4.0 29B · kilo** xingchen-agi/xing4.0-29b-a4b:free	kilo	262k	236k	—	Sparse 29B MoE: cheap long-output work — bulk rewrites and data generation.
25	**Nemotron 3 Nano Omni · kilo** nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free	kilo	256k	65k	✓	Small omni reasoning + vision; thrifty for image-aware batch jobs.
26	**Nemotron 3.5 Content Safety · kilo** nvidia/nemotron-3.5-content-safety:free	kilo	128k	8k	✓	Moderation and safety classification — policy checks, not creative work.

3 Field notes — what each model is for

00 · Auto Combo combo
Chains free links and only fails over when a link dies before producing content. For unattended jobs and a "just work" default.

01 · DeepSeek V4.1 Flash free
All-round reasoning and coding with the biggest output budget here (384k). The main driver for real coding sessions and long writes.

02 · GLM 5.3 Flash free
Fast, well-structured, strong multilingual coverage, vision. A second opinion, translations, quick screenshot reviews.

03 · MiMo V2.6 Flash free
Balanced multimodal generalist. UI iteration, screenshot → notes, everyday image + text chat.

04 · Nemotron 3.5 Super VL free
Reads images carefully — charts, tables, UI, OCR-style extraction. Diagram → spec, screenshot → bug report.

05 · Nemotron 3.5 Lightning free
The quickest free-lane feel on short context. Quick chat, small edits, tight tool loops (128k cap).

06 · Kilo Auto kilo
Routes inside Kilo's free pool with the leanest Kilo overhead. Hands-off default; prompts may be logged upstream.

07 · Nemotron 3 Ultra 550B kilo
The heaviest reasoning in the stack. Architecture calls, postmortems, hard debugging — slow, so not for churn.

08 · Nemotron 3.5 Super VL (Kilo) kilo
Same careful vision work without the Zen-lane prompt overhead. Image-heavy tasks when the free lane throttles.

09 · Nemotron 3.5 Lightning (Kilo) kilo
1M context at speed. First-pass triage of huge logs or diffs, then hand the finding to a deeper model (65k out).

10 · Inkling kilo
Multi-step agentic work with 262k outputs. Autonomous refactor loops and long tool-driven research.

11 · Inkling Small kilo
Same envelope at lower cost per iteration. test-fix-test cycles; escalate to 10 for the final pass.

12 · Dots3 Note Preview kilo
Document scale in *and* out — a 460k output ceiling. Whole books, reports, specs; long inputs restructured.

13 · Glyph Cluster kilo
Mass transformation: rewrite, translate, summarize large inputs with room to write back. Not a deep reasoner.

14 · Flint 1.0 kilo
Reasoning plus long output. Design docs and thorough reviews that need depth *and* length.

15 · North Mini Code kilo
Code review, refactors, tests, diffs. Purpose-built; pair with 01 when the task is half prose.

16 · OpenRouter Free Router kilo
Breadth across OpenRouter's own `:free` catalog. Variety and fallback when Kilo's pool is saturated.

17 · Gemma 4 31B kilo
Small-model discipline: predictable formatting, multilingual, vision. Image triage at volume.

18 · Big Pickle free
Roomy general chat with vision. Everyday Q&A, doc drafting, a comfortable fallback when flagships throttle.

19 · Beam free
Dependable mid-tier reasoning. A safe, consistent everyday choice rather than a specialist.

20 · Beam 1.0 (Kilo) kilo
Reasoning at 1M context with leaner overhead. Long analysis when the free lane is busy.

21 · Flint 40B free
40B open-model judgment. A second perspective on plans and mid-size engineering questions.

22 · Pulse Alpha free
A preview of upcoming free frontier models. Experimentation only — expect volatility.

23 · LFM2.5 2.6B kilo
Tiny and fast: titles, tags, classification, extraction, routing. A utility model, not a reasoner.

24 · Xing4.0 29B kilo
Cheap long-output generation via sparse MoE. Bulk data, mass rewrites and synthetic examples at scale.

25 · Nemotron 3 Nano Omni kilo
Small omni reasoning plus vision. Alt text, screenshot triage at volume, image-aware batch jobs.

26 · Nemotron Content Safety kilo
Safety and moderation classification. Policy checks and filtering — not a chat model.

4 Pick by task

IF YOU NEED…	USE	WHY
A default that just works	01 DeepSeek V4.1 Flash, or 00 combo when flaky	Best balance of reasoning, speed and 384k output
The hardest thinking	07 Nemotron 3 Ultra 550B, then 01	Largest model in the stack, 1M context
Screenshots, UI, charts, OCR	04 (free lane) or 08 (kilo VL)	Vision-language specialists with 1M context
Whole repo / book in one shot	12 Dots3 Note, 09 Lightning, 13 Glyph	512k–1M input with up to 460k output
Coding & refactors	15 North Mini Code, 01, 14 Flint 1.0	Purpose-built code models + long-output generalists
Agentic tool loops	10 Inkling, 11 Inkling Small, 06 Kilo Auto	1M context, vision, 262k outputs
Bulk extraction / classification	23 LFM2.5 2.6B, 24 Xing4.0, 26 Content Safety	Cheap, fast, purpose-shaped
Variety when Kilo is busy	16 OpenRouter Free, 00 combo	Routers over independent pools

5 Measured latency & cost profile

LANE	OBSERVED	PROMPT OVERHEAD
00 combo	1.0–2.5 s	~5.9k tok
01 DeepSeek V4.1 Flash	1.2–1.4 s	~5.9k tok
06 Kilo Auto	1.1 s (direct lane)	~2k tok
fcc / OpenRouter `:free`	2.2–5.1 s	~22 tok

Reading the numbers. Free-lane models are fastest but carry a large injected system prompt (that's the cost of keyless frontier access); Kilo lanes are leaner; FCC via OpenRouter is the leanest and slowest. A one-line question costs ~5.9k tokens on the free lane — for cost-sensitive bulk work prefer 23/24/26 or the `fcc` provider.

Ranking caveat. These are latency and size facts plus practical fit. No benchmark scores are implied; upstream rosters change and quotas rotate. Re-run `npm run bakeoff` any time for a fresh scorecard.

6 Using them in OpenCode

- **Select:** `/models` → provider **00-free** → `NN-name`. The numbers are the ranking.
- **Refresh the roster** after upstream changes: `npm run sync` (re-pulls live models, keeps this order).
- **Reorder the top**: edit the `PRIORITY` array in `bin/sync-models.mjs`, then `npm run sync`. Unlisted new models append alphabetically — nothing disappears.
- **Privacy:** Kilo-lane prompts may be logged upstream; route anything sensitive through your own aggregate keys (dashboard → Aggregate) or local Ollama via the `fcc` provider.
- **More lanes** (dashboard at `http://127.0.0.1:18900`): EAC (GitHub sign-in + star) and account channels (LobsterAI, Qoder, TRAE, Cline, Loomy, Raccoon, MiniMax Code, ZCode, Gemini) add models after you sign in.